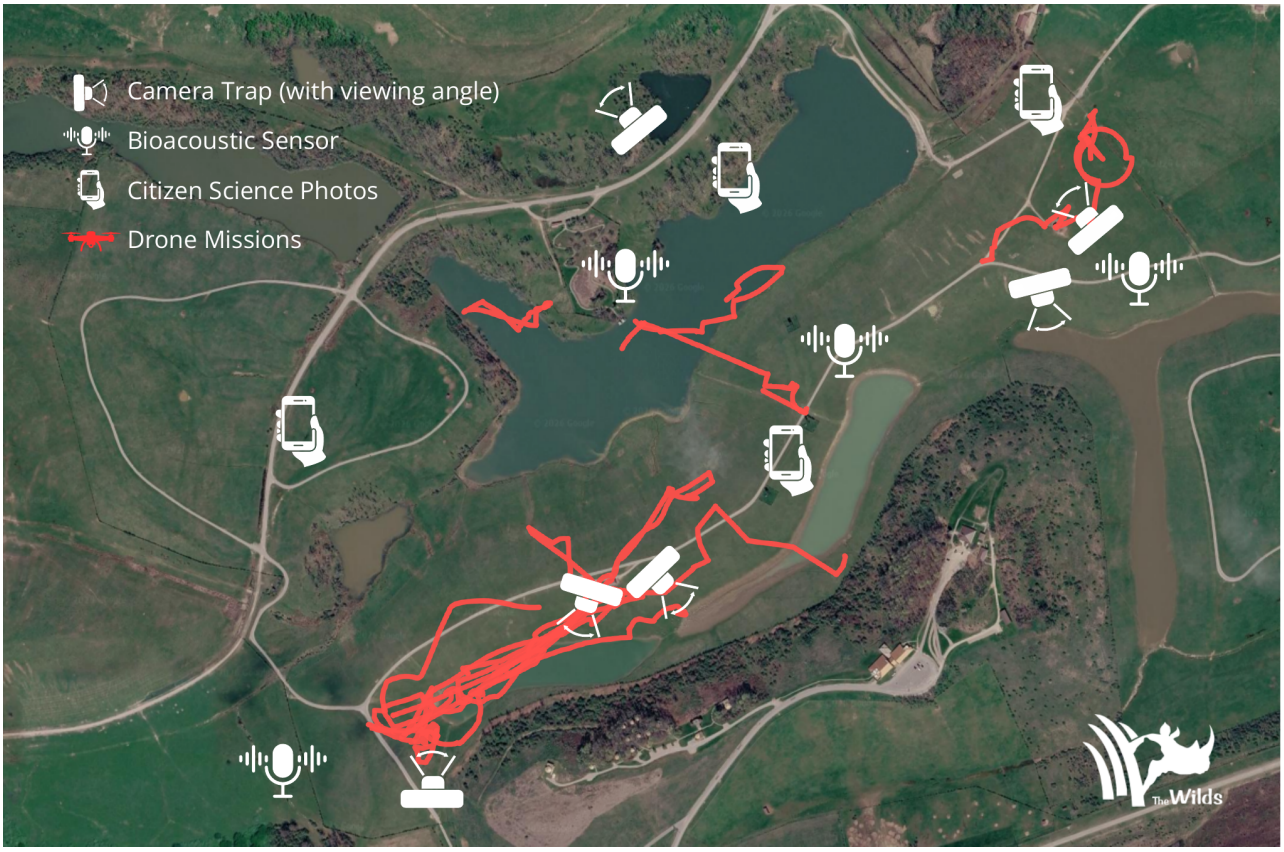
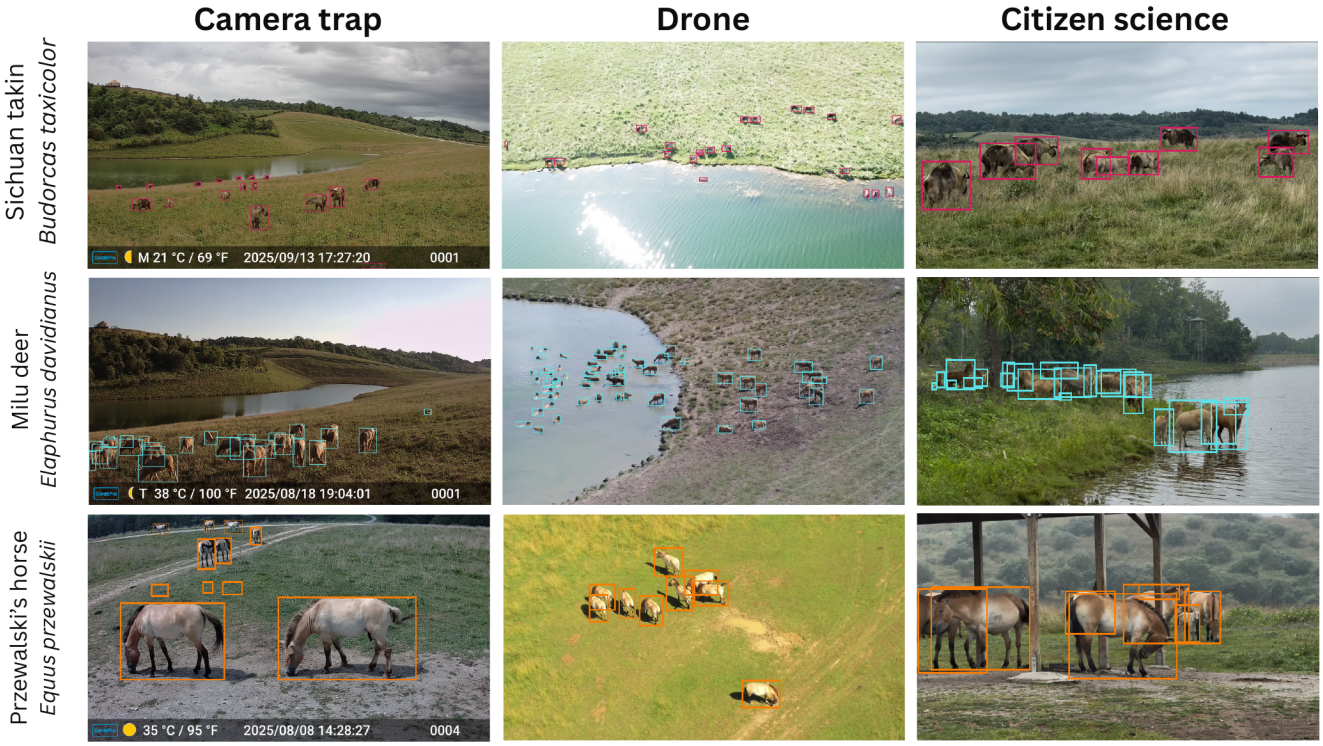
smartwilds — dataset card (review draft)

Rendered from release/hf_smartwilds_20261022/smartwilds/README.md

Dataset Card for SmartWilds



Sensor deployment across The Wilds: camera traps with viewing angles, Song Meter recorders, citizen-science photo locations, and drone flight tracks. Satellite imagery © Google.



The same three managed focal species (Sichuan takin, Milu deer, Przewalski's horse) observed by camera trap, drone, and citizen-science photography.

> **Status: staging incomplete.** The merged Event table is owner-only on the cluster. Rebuild with `release/scripts/build_hf_release.py` once it is readable.

SmartWilds is a multi-modal sensing dataset from three 2025 field campaigns at The Wilds (Cumberland, Ohio). Each modality has its own dataset and DOI:

Dataset	Content
smartwilds-camera-traps	235,778 images, 5,113 clips, 224,149 Occurrences
smartwilds-bioacoustics	2,371 WAV recordings, 9,960 Occurrences
smartwilds-drones	39 videos, 15 flight sessions, 318,203 Occurrences
smartwilds-citizen-science	1,103 media files, 622 Occurrences
smartwilds (this dataset)	study root, merged Event table, text records, arrival traces

All five datasets share one Event hierarchy:

```
TheWilds_2025 (study, 2025-06-30/2025-09-08)
└── TheWilds_2025_<modality> (Project)
    └── SensorDeployment / flight session / observation session
        └── AudioRecording | MediaEvent
```

Contents

- `data/events_all_modalities.csv`: all 2,470 Events. A user can join Occurrences from any modality dataset to it on `eventID`.
- `data/text_*.csv`, `data/site_selection_rationale.csv`: 30 text-derived Occurrences, plus the reasoning behind where each sensor was placed.

- `metadata/sensor_configurations.md`: per-device configurations for every modality.
- `analysis/arrival_traces_v2/`: Campaign-2 arrival streams at two layers.
  - Acquisition: one WAV recording per arrival.
  - Detection: one machine Occurrence per arrival.

  Each stream comes with source-record IDs, inclusion rules, and checksums. The camera image-capture acquisition stream can be rebuilt from `smartwilds-camera-traps/data/images.csv`.
- `SUPPLEMENTAL.md`: the paper's supplemental material — extended glossary, per-deployment inventories, all 18 timestamp-repair methods, drone ground projection, the standards crosswalk, typed missingness, model accuracy, validation checks, the field dictionary, arrival statistics, reproduction commands, and the full defect inventory.
- `metadata/DEFECT_INVENTORY.md` (and `defect_inventory.csv`): all 20 defect classes found during integration, each with its scope, what detected it, the repair applied, and the residual risk.
- `metadata/provenance/`: the final release audit.

Citation

Cite the paper, plus each modality dataset you use. **[TODO: DOIs]**

Dataset Card Contact

Jenna M. Kline (kline.377@osu.edu)

Appendix: Hugging Face metadata (YAML front matter)

```yaml
license: cc0-1.0
language:
- en
pretty_name: SmartWilds (The Wilds 2025) - Study Hierarchy, Text Records, and Workload Traces
tags:
- biology
- ecology
- biodiversity
- multimodal
- sensor-networks
- darwin-core
- humboldt-eco
- fair2
- smartwilds
- the-wilds
size_categories:
- 100K<n<1M
description: >-
  The study-level component of the SmartWilds collection: the merged Darwin
  Core Event hierarchy (2,470 Events) rooted at TheWilds_2025, field text
  records and site-selection rationale, sensor configurations, and versioned
  Campaign-2 arrival traces for workload characterization.
```